

Introduction to Complex Analysis
MATH 4230-40

Name (Print): _____

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- This exam contains 6 pages (including the front and back of this cover page) and 5 problems. Check to see if any pages are missing.
 - This is the third exam and will count for 20% of your final grade. You have until Friday, May 13th at 11:59 PM to complete this exam. You may use consult your notes and textbook during this exam, but no other outside sources may be consulted.
 - There are 140 points total in this exam.

The following rules apply:

- Every problem on this exam requires work of some form. An incorrect answer supported by substantially correct calculations and explanations will receive partial credit. **A correct answer that has no supporting work will not receive full credit.**
- Organize your work, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little to no credit.

Page:	2	3	4	5	6	Total
Points:	20	20	60	20	20	140
Score:						

1. 20 points Let $\mathcal{H}$ denote the upper half plane; i.e., $\mathcal{H} = \{z \in \mathbb{C} : \operatorname{Im} z > 0\}$. Suppose that $a, b, c, d \in \mathbb{R}$ are such that $ad > bc$ and let T be the fractional linear transformation

$$T(z) = \frac{az + b}{cz + d}.$$

Show that $T(\mathcal{H}) = \mathcal{H}$.

2. 20 points The *cross ratio* of four distinct points z_1, z_2, z_3, z_4 is defined by

$$[z_1, z_2, z_3, z_4] = \frac{z_4 - z_1}{z_4 - z_2} \frac{z_3 - z_2}{z_3 - z_1}.$$

Without using Proposition 5.2.4, show that every fractional linear transformation T has the property that

$$[T(z_1), T(z_2), T(z_3), T(z_4)] = [z_1, z_2, z_3, z_4].$$

3. This question concerns the *fixed points* of a fractional linear transformation T (that is, points z for which $T(z) = z$).
- (a) 20 points Let T be a fractional linear transformation that is not the identity map. Show that T has at most two fixed points.
 - (b) 20 points Give an example of a fractional linear transformation that has no fixed points.
 - (c) 20 points Find the fixed points of $T(z) = \frac{z}{z+1}$.

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4. 20 points Let T and R be fractional linear transformations. Prove that $T \circ R$ is a fractional linear transformation as well.

5. 20 points Write a paragraph explaining intuitively what is "actually" meant by the statement

$$1 + 2 + 3 + \dots = -\frac{1}{12}$$

in terms of analytic continuation and the Riemann zeta function.